

4.2 Kinematics of motion in a straight line

Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Velocity is derivative of displacement; acceleration is derivative of velocity. Integrate acceleration to velocity and velocity to displacement with constants fixed by conditions. Constant-acceleration equations require truly constant a ; sign conventions must be maintained in multi-stage travel.

Worked example: follow the reasoning

For $s=t^3-3t$, $v=3t^2-3$ and $a=6t$; at $t=1$ $s=-2$ and $v=0$.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A particle has $v=4t-2$ m s⁻¹ and $s(0)=3$ m. Find $s(t)$ and displacement from $t=0$ to 2. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Integrate: $s=2t^2-2t+C$, $C=3$ [2]; $s(2)-s(0)=4$ m [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.